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 Oefeningenreeks 3G nr. 18.1

$$f(x) = x\sqrt{x}$$

dom, Im, periode, snijpunten assen

$$\text{dom } f = \mathbb{R}^+, \text{Im } f = \mathbb{R}^+, \text{per. } f = /$$

$$0 = x\sqrt{x} \Leftrightarrow x = 0 \text{ of } \sqrt{x} = 0$$

Snijpunt(en) x-as: $(0, 0)$, Snijpunt Y-as: $(0, 0)$

Asymptoten:

VA: Geen VA mogelijk

$$\text{Ha: } \lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} x = +\infty \Rightarrow \text{Geen HA}$$

$$\lim_{x \rightarrow -\infty} f(x) = \text{bestaat niet}$$

$$\text{SA: } \lim_{x \rightarrow +\infty} \frac{f(x)}{x} = \lim_{x \rightarrow +\infty} \frac{x\sqrt{x}}{x} = +\infty \Rightarrow \text{Geen SA}$$

Afgeleiden:

$$\frac{df}{dx}(x) = x \cdot \frac{1}{2\sqrt{x}} + \sqrt{x} = \frac{x}{2\sqrt{x}} + \frac{2x}{2\sqrt{x}} = \frac{3x\sqrt{x}}{2x} = \frac{3\sqrt{x}}{2}$$

$$\frac{d^2f}{dx^2}(x) = \frac{3}{2} \cdot \frac{1}{2\sqrt{x}} = \frac{3}{4\sqrt{x}}$$

x	$-\infty$	0	$+\infty$	
$f'(x)$	///	0	$+$	$+$
$f''(x)$	///	$ $	$+$	$+$
$f(x)$	///	0 $m(0)$	$+$ $\nearrow$ $\smile$	$+$ $\nearrow$ $\smile$

References

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